Mental Health
Risk Prediction

Using CDC BRFSS 2024 Survey Data

Section A: Non-Technical Summary

Section B: Technical Overview

457,670	37	5
Respondents	Variables	Models Tested

OUR TEAM

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The Problem — Why Does Mental Health Matter?

1 in 5

U.S. adults experience mental illness each year

15%

of our 457K respondents report frequent mental distress

$280B

estimated annual cost of mental illness in the U.S.

Core Question

Can we use everyday survey data to predict who is most at risk — **BEFORE** they reach a crisis?

Early Detection

Identify high-risk individuals before mental health deteriorates to crisis level

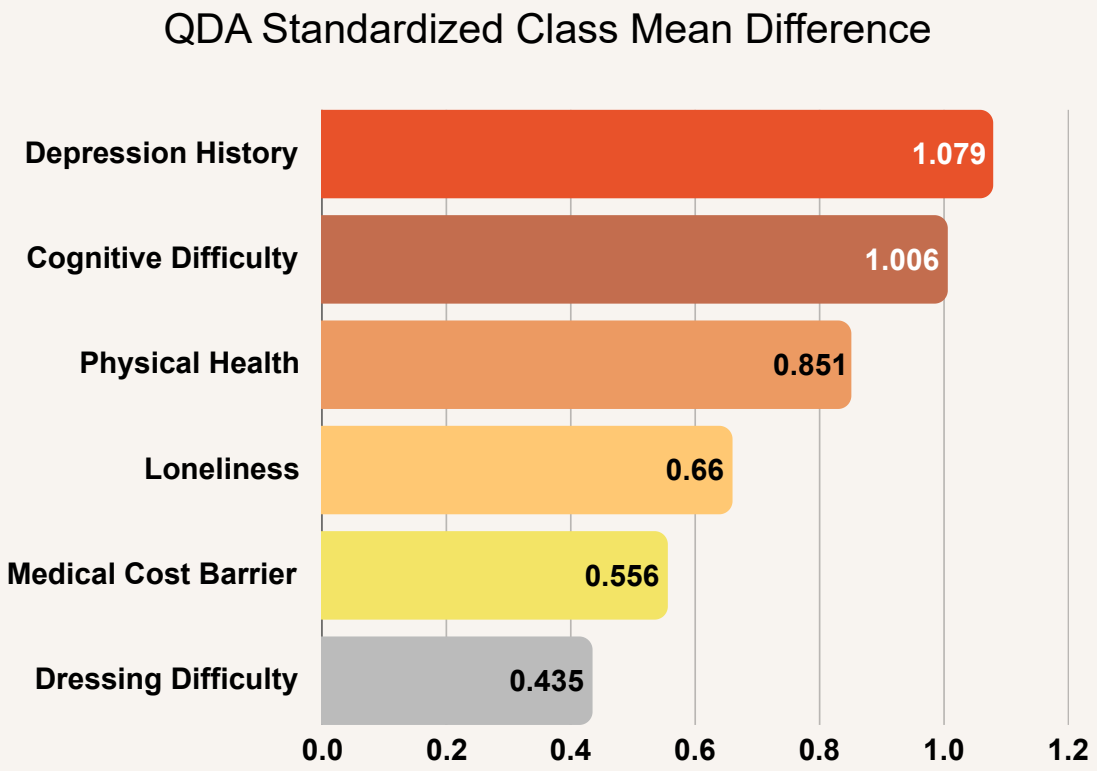
Resource Allocation

Direct mental health services toward groups who need them most efficiently

Policy Guidance

Give governments and insurers evidence to design better support programs

What Did We Find? —The 3 Biggest Predictors



#1

Depression History 1.08

Having ever been diagnosed with depression is by far the single strongest predictor.

#2

Cognitive Difficulty 1.01

People who struggle to concentrate or make decisions report far more bad mental health days.

#3

Physical Health 0.85

Body and mind are deeply linked — poor physical health days strongly predict poor mental health days.

What Do We Recommend?

For business leaders, healthcare providers, and policymakers

Healthcare Providers

Screen for Depression History First

Depression History alone accounts for 28% of predictive power. Any mental health screening program should make depression history the primary intake question.

Policymakers

Remove Financial Barriers to Care

People who can't afford doctor visits show significantly worse mental health outcomes. Subsidized access programs would directly reduce risk for the highest-need population.

Government & Insurers

Target Divorced & Low-Income Groups

EDA shows divorced, separated, and low-income individuals consistently rank as high-risk. Targeted outreach and community support programs could intervene early.

Healthcare Systems

Treat Physical & Mental Health Together

Physical health is the 3rd strongest predictor. Integrated care models that treat body and mind together will be more effective than siloed approaches.

Note: Our best model (Random Forest) explains ~33% of mental health variation. These are strong signals, not certainties — human judgment remains essential.

SECTION B

Technical Overview

1 Data & Preprocessing

2 Interesting Variables from EDA

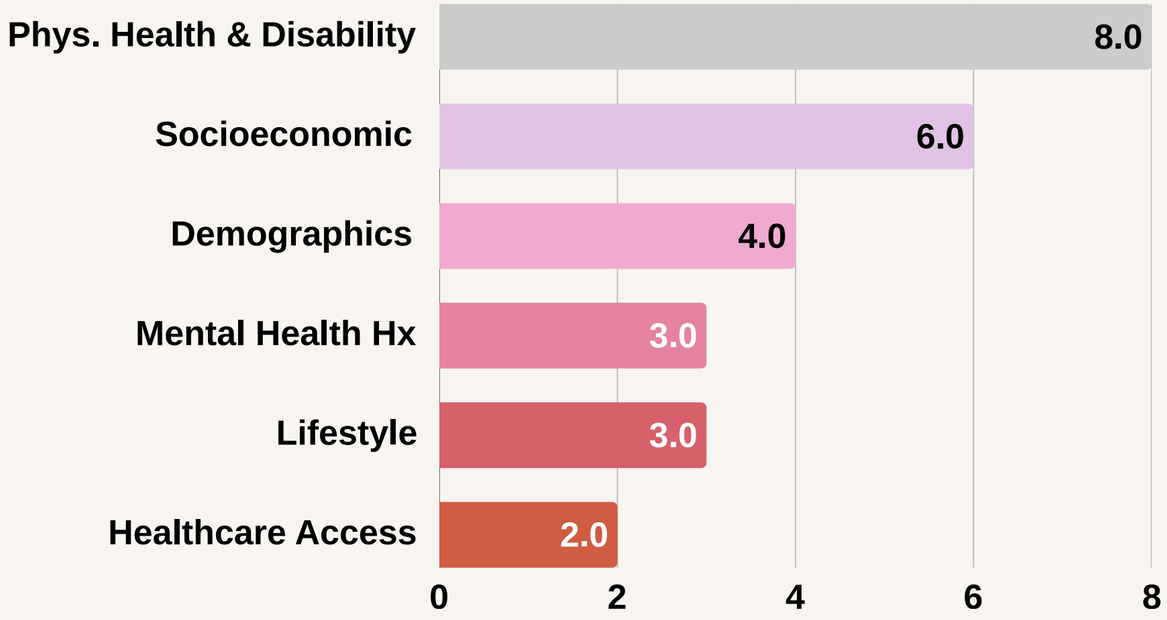
3 EDA Findings

4 Models & Metrics

5 Results

Data — CDC BRFSS 2024

Behavioral Risk Factor Surveillance System · 457,670 respondents · 301 variables → 37 selected



Key Preprocessing Steps

01 Survey Code Recoding
77=Don't know, 99=Refused → NaN
88=None → 0 · Binary 2=No → 0

02 Column Removal
>200K missing (ACE series) → dropped
BMI engineered but dropped (r≈0.04)

03 Outlier Treatment
HEIGHT >7'11" → removed
WEIGHT ≥9000 (codes) → removed
0 duplicate rows found

Scenario A — Classification

Y = frequent_distress (1 if ≥14 bad mental health days, else 0) · 75/25 stratified split

Scenario B — Regression

Y = Mental Health (continuous 0–30 days of poor mental health) · 80/20 random split

Data — EDA

Interesting Finding in EDA

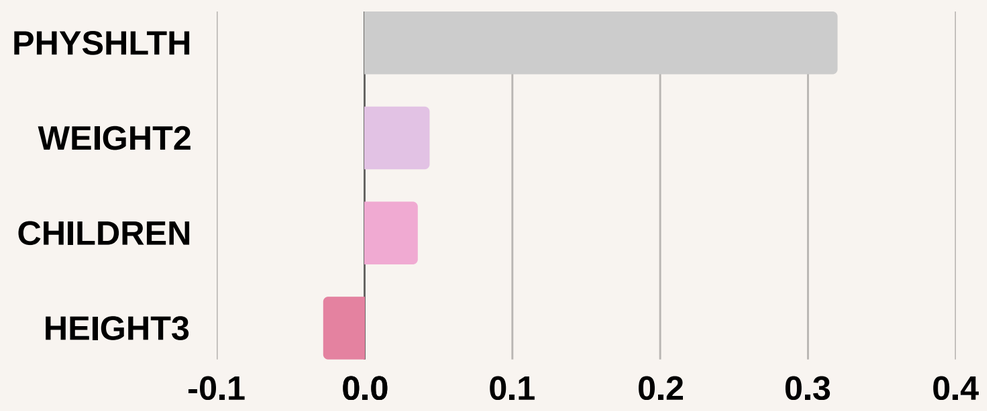
Smoker
+ Can't Afford Dr.
= worst combo

Divorced women
> Divorced men

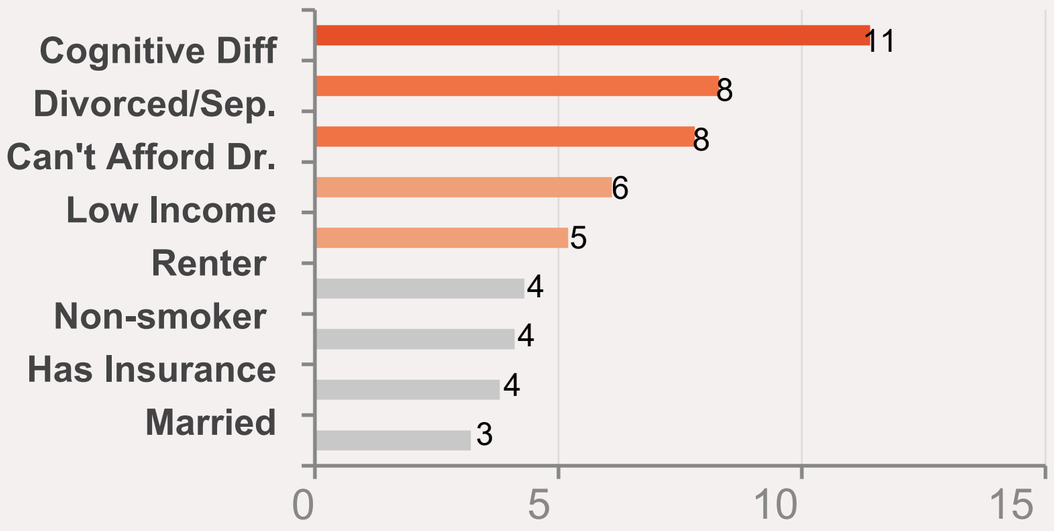
Physical health vs
Mental health scatter

EDA — Data Patterns & High-Risk Groups

Continuous Variable Correlations with MENTHLTH



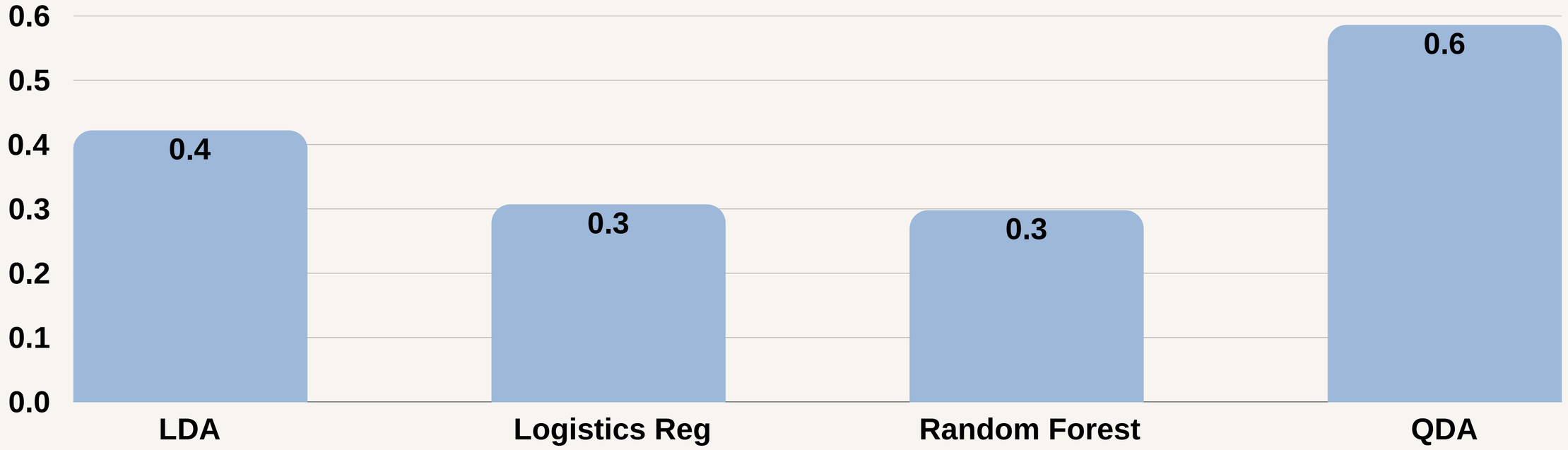
Avg Poor Mental Health Days by Group



3 Key EDA Takeaways

- Physical Health is the only continuous variable with meaningful correlation (r=0.32)
- Cognitive Difficulty group averages 11.4 days — 3.5× higher than the married group (3.2)
- 85/15 class imbalance → Accuracy is misleading; Precision-Recall metrics required

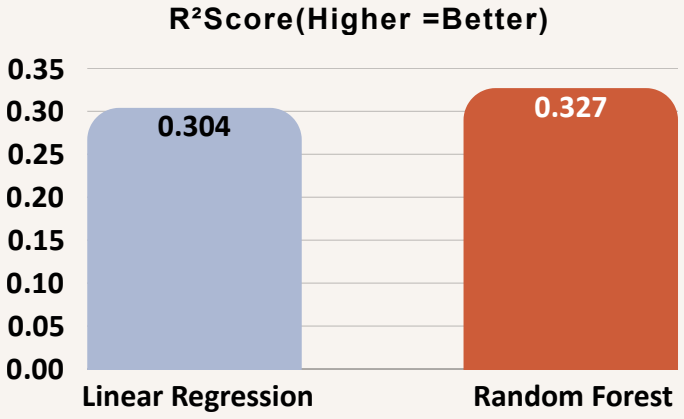
Results — Scenario A (Classification)

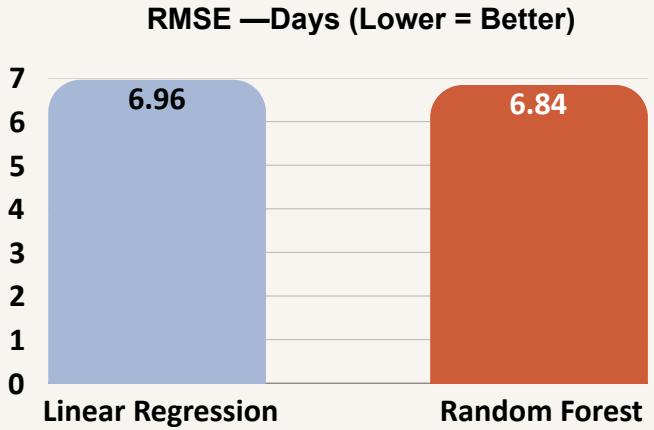


Model	Recall (Class 1)	ROC AUC	Precision (Class 1)	F1 (Class 1)	Best For
LDA	0.422	—	0.569	0.485	Baseline
QDA ★	0.586 ★	0.803	0.374	0.457	Best Recall — fewest missed cases
Logistic Reg.	0.307	0.837	0.650 ★	0.417	Best precision at threshold
Random Forest	0.298	0.846 ★	0.666	0.412	Best AUC — most reliable overall

Results —Scenario B (Regression)

Predicting the number of poor mental health days(0–30) · Metric:R^2 and RMSE





Interpreting the Results

- R^2 = 0.327 vs 0.100: Random Forest explains 3×more variance than Linear Regression —non-linear interactions between variables matter.
- RMSE = 6.84 day.
- **Random Forest captures complex variable interactions that a linear model simply cannot —justifying the added complexity.**

Model Selection & Findings

Models Tested

LDA	Logistic Reg.	Linear Reg.	Random Forest	QDA
Scenario A · baseline	Scenario A · baseline	Scenario B · baseline	Scenario A + B	Scenario A · selected

What We Found

Top 3 predictors

Depression · Cognitive Diff · PhysHealth

Confirmed by both QDA class means and RF feature importance

Precision-Recall tradeoff

QDA: best recall, lower precision

RF: lower recall, best AUC

R^2 ~0.30 (both models)

Mental health is genuinely complex
~33% variance explained by the model.

Thank You

3 Key Takeaways

01

Depression History Dominates

We prefer QDA for its 58.6% recall, driven mainly by depression history, cognitive difficulty, poor physical health, difficulty with daily tasks, and limited access to healthcare.

02

QDA Wins —But Mental Health Remains Complex

We prefer QDA for its high recall BUT:

*A reminder that prediction models complement, but cannot replace, human judgment.

03

Data tells the Real Story

Mental distress is not an isolated issue, but part of a broader pattern of overall health and socioeconomic vulnerability.